Block 1 Technical Report

Operational Transconductance Amplifier (OTA)

1.8-V Fully Differential OTA for a Switched-Capacitor Delta-Sigma Modulator

SKY130 transistor-level design and ngspice characterization

Final status. Nominal schematic-level characterization is complete. The OTA meets the selected nominal gain/stability target at TT, 1.8 V, 27 °C. PVT stress testing identifies a major fixed-bias robustness limitation; therefore this block is *characterized* but is not yet PVT-qualified or tapeout-ready.

Finalized characterization report

Revision 2 — 2 November 2026

Contents

1 Executive Summary **2**

2 Design Context and Objectives **2**

3 Transistor-Level Architecture **3**

3.1 Initial baseline schematic 3

3.2 Final characterized architecture 4

4 Compensation Selection and AC Characterization **4**

5 Closed-Loop Small-Signal Transient **5**

6 Large-Signal Slew Rate **6**

7 Quiescent Current and Power **7**

8 Noise Characterization **8**

9 CMRR and PSRR **9**

10 PVT Stress Characterization **9**

10.1 Method 9

10.2 Bias-sweep diagnostic 11

11 Revision 2 Addendum: Compensation Sensitivity and Integration Notes **12**

11.1 Compensation capacitance: MF=80 versus MF=100 12

11.2 Relation to the OTA used in the modulator netlists 12

12 Final Performance Summary **13**

13 Design Assessment and Block Closure **14**

A Reproducibility Notes **14**

Revision History

Rev.	Date	Change
1	–	Finalized nominal characterization and fixed-bias PVT stress test.
2	2 Nov 2026	Resolved the MIM multiplicity note (Section 3.2); added the compensation-sensitivity and integration addendum (Section 11). No nominal metric of Revision 1 was changed.

1 Executive Summary

This report closes the first analog building block of the mixed-signal portfolio: a fully differential operational transconductance amplifier implemented with SKY130 1.8-V MOS devices. The design was characterized at transistor level using Xschem/ngspice. The nominal design uses direct MIM-capacitor compensation with an instantiated compensation setting of `MF=80` and two 20 pF external output loads in the testbench.

At the nominal TT / 1.8 V / 27 °C point, the OTA achieves 69.34 dB DC gain, a 3.153 12 MHz unity-gain frequency (UGF), and 61.3704° phase margin. Closed-loop small-signal testing gives a 56.19 ns 10–90% rise time, 186.86 ns settling to 1%, and 294.80 ns settling to 0.1%. Large-signal testing gives approximately 5.06 V/µs positive and 4.97 V/µs negative slew rate. Nominal DC power is 11.59 mW.

Noise, common-mode rejection, and supply-rejection tests were also performed. The integrated input-referred noise from 1 Hz to 25 kHz is approximately 42.07 μV_{rms}. Extremely high nominal CMRR and PSRR values are interpreted cautiously because perfect schematic symmetry suppresses differential error and can push rejection measurements toward the numerical floor.

The most important design limitation appears in PVT stress tests. With the fixed nominal bias voltage of 0.507 621 V, the operating point collapses at supply, temperature, and process extremes. This result identifies the next engineering requirement clearly: a PVT-aware bias/current-reference strategy is needed before the OTA can be considered robust for silicon implementation.

2 Design Context and Objectives

The OTA is intended as an analog core for a second-order switched-capacitor delta-sigma modulator with a nominal sampling frequency of 4 MHz and an oversampling ratio of 80. The corresponding nominal signal band used for noise integration is

$$BW = \frac{f_s}{2\,OSR} = \frac{4\,\text{MHz}}{160} = 25\,\text{kHz}. \tag{1}$$

The characterization priorities were therefore: sufficient low-frequency gain, stable closed-loop behavior, useful settling speed relative to the sampling period, balanced large-signal response, quantifiable noise, and an explicit assessment of PVT robustness.

The nominal electrical conditions used throughout the final characterization are summarized in Table 1.

Table 1: Nominal characterization conditions.

Parameter	Value	Note
Supply voltage V_{DD}	1.8 V	SKY130 1.8-V devices
Input common mode	0.4 V	$V_{in,p} = V_{in,n}$ in DC
Bias voltage V_{bias}	0.507 621 V	Fixed nominal bias
CMFB control V_{cmfb1}	0.9 V	External testbench bias
Output load	20 pF/side	External load capacitor
Process/temperature	TT / 27 °C	Nominal corner

3 Transistor-Level Architecture

3.1 Initial baseline schematic

The design process began from the original fully differential OTA schematic shown in Figure 1. This baseline implementation included the PMOS differential input pair, cascaded/current-mirror gain paths, complementary output devices, the common-mode-control transistor, and two V_{bz}-controlled compensation branches (`Vbz_L` and `Vbz_R`) connected around the output-side compensation network. In the initial schematic, the MIM compensation capacitors were explored with a much larger scaling setting (shown in the captured design as `MF=1000`).

This baseline is intentionally retained in the report because it documents the actual starting point of the design, not only the final successful configuration. The subsequent characterization showed that the original compensation/nulling arrangement did not provide the desired stability-performance trade-off. The design was therefore simplified into the final `OTA_experiment` variant, in which the nulling branches were removed and the direct-C compensation factor was retuned to `MF=80`.

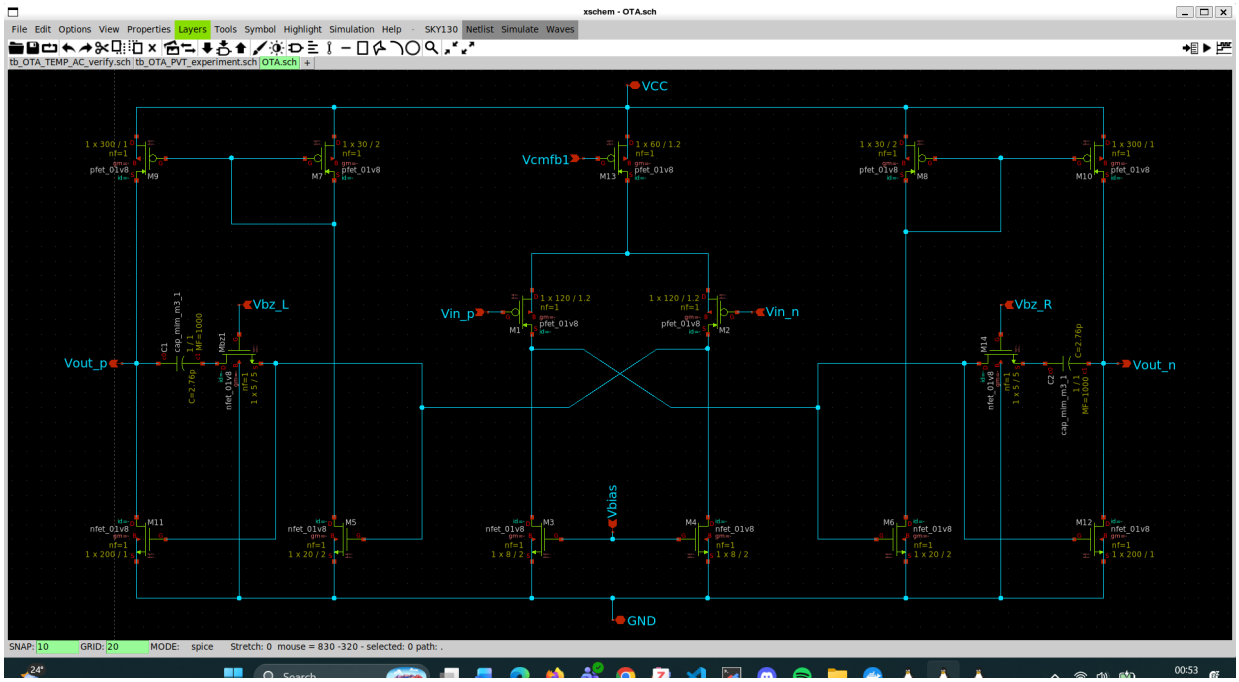


Figure 1: Initial baseline OTA schematic in Xschem before the direct-C compensation simplification. The V_{bz}-controlled branches and the original large compensation-capacitor setting are preserved here as part of the design history.

3.2 Final characterized architecture

The final experimental OTA contains a PMOS differential input stage, internal gain/current-mirror devices, complementary output devices, and a PMOS common-mode-control device. Direct compensation capacitors are connected from the two outputs to the corresponding internal high-impedance nodes. The earlier nulling-branch experiment was removed from the final experimental schematic.

Table 2: Final transistor sizing used in `OTA_experiment`.

Devices	Type	W (µm)	L (µm)
M1, M2	PMOS	120	1.2
M3, M4	NMOS	8	2
M5, M6	NMOS	20	2
M7, M8	PMOS	30	2
M9, M10	PMOS	300	1
M11, M12	NMOS	200	1
M13	PMOS	60	1.2

The final compensation configuration uses two SKY130 MIM capacitors with schematic parameter `MF=80`. **Multiplicity note resolved (Revision 2).** A dedicated AC capacitance test compared the Xschem MIM symbol with an ideal 1 pF reference capacitor. The symbol emits both `MF=N` and `m=N` in the netlist; the SKY130 MIM model ignores `MF`, and only `m` multiplies the capacitance. One unit ($W = L = 1$ with the netlist scale of 10^{-6}) measures 2.645 fF. Consequently `MF=80` is 211.6 fF per compensation capacitor, `MF=100` is 264.5 fF and `MF=1000` is 2.645 pF, and no double multiplicity occurs. The calibration reference was recovered as 1000.00 fF. This is a nominal-model measurement (status: measured, not final).

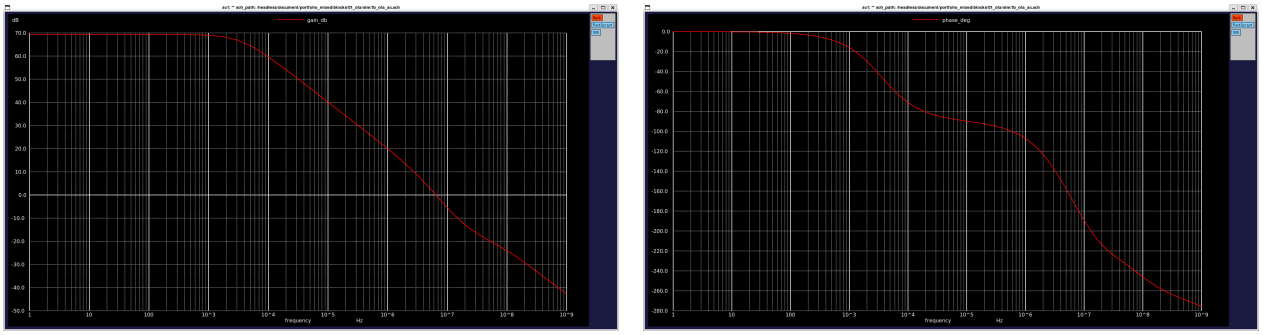
4 Compensation Selection and AC Characterization

Initial compensation experiments produced inadequate phase margin. A direct-capacitor approach was then explored by increasing the effective MIM-capacitor scaling. The final selected point was the first tested configuration that provided approximately 60° nominal phase margin while retaining MHz-class bandwidth.

For differential AC excitation,

$$A_v(f) = \frac{V_{out,p} - V_{out,n}}{V_{in,p} - V_{in,n}}. \tag{2}$$

The sign was normalized in the loop-gain post-processing so that phase margin could be read consistently at the 0-dB crossing.



(a) Differential gain magnitude.

(b) Unwrapped differential phase.

Figure 2: Nominal AC response of the selected direct-C configuration.

The accepted nominal AC metrics are

$$A_0 \approx 69.34\,\mathrm{dB}, \tag{3}$$

$$f_{UGF} \approx 3.153\,12\,\mathrm{MHz}, \tag{4}$$

$$\phi(f_{UGF}) \approx -118.630^\circ, \tag{5}$$

$$PM = 180^\circ + \phi(f_{UGF}) \approx 61.3704^\circ. \tag{6}$$

These results establish a stable nominal small-signal operating point for the subsequent closed-loop characterization.

5 Closed-Loop Small-Signal Transient

A numerically isolated behavioral feedback harness was used to preserve an input common-mode voltage near 0.4 V while closing the differential loop. A differential reference step of approximately 0 to 50 mV was applied. The testbench was first validated by checking the initial transient solution before measurement expressions were enabled.

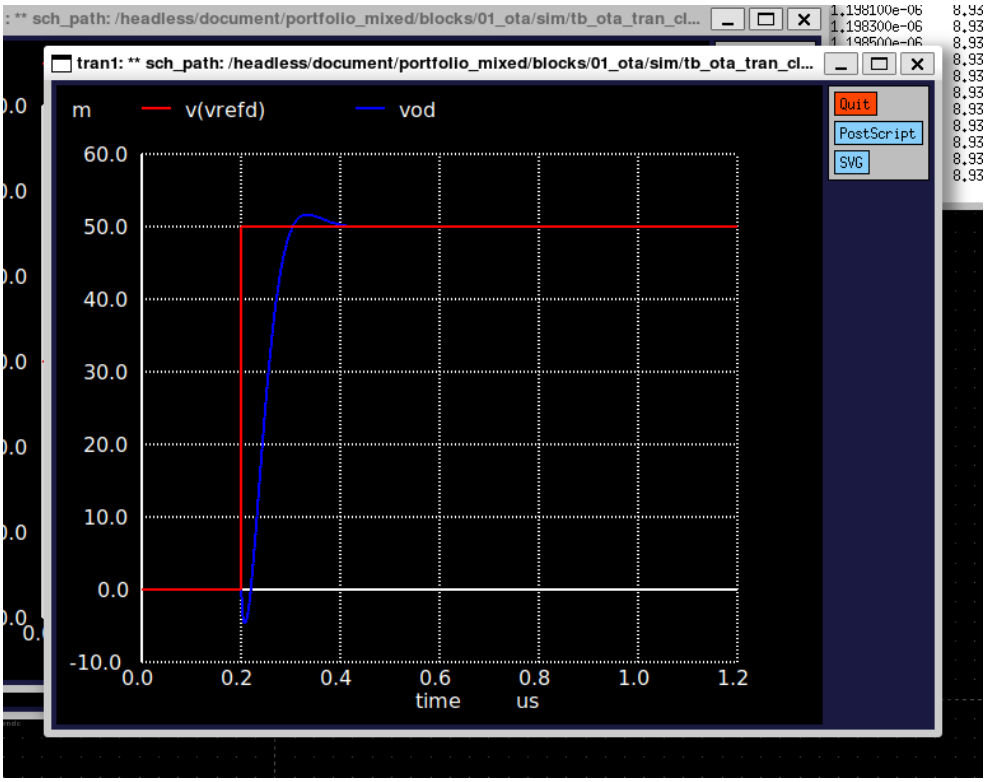


Figure 3: Closed-loop 50 mV differential step response.

The measured results are listed in Table 3.

Table 3: Closed-loop small-signal transient results.

Metric	Result
10–90% rise time	56.1867 ns
Settling to 1%	186.859 ns
Settling to 0.1%	294.795 ns
Overshoot	3.570 43 %
Output common-mode droop	2.3165 mV
Final differential output	49.893 mV (approximately)

For a 4 MHz sampling clock, one full period is 250 ns. The measured 1% settling time is below one full clock period, while the 0.1% settling time exceeds it. A two-phase switched-capacitor implementation may provide an even tighter effective settling window, so final system-level integration must reassess this timing budget.

6 Large-Signal Slew Rate

A larger differential closed-loop reference step of approximately 0 to 300 mV was used to expose large-signal charging/discharging behavior. The validated response is shown in Figure 4.

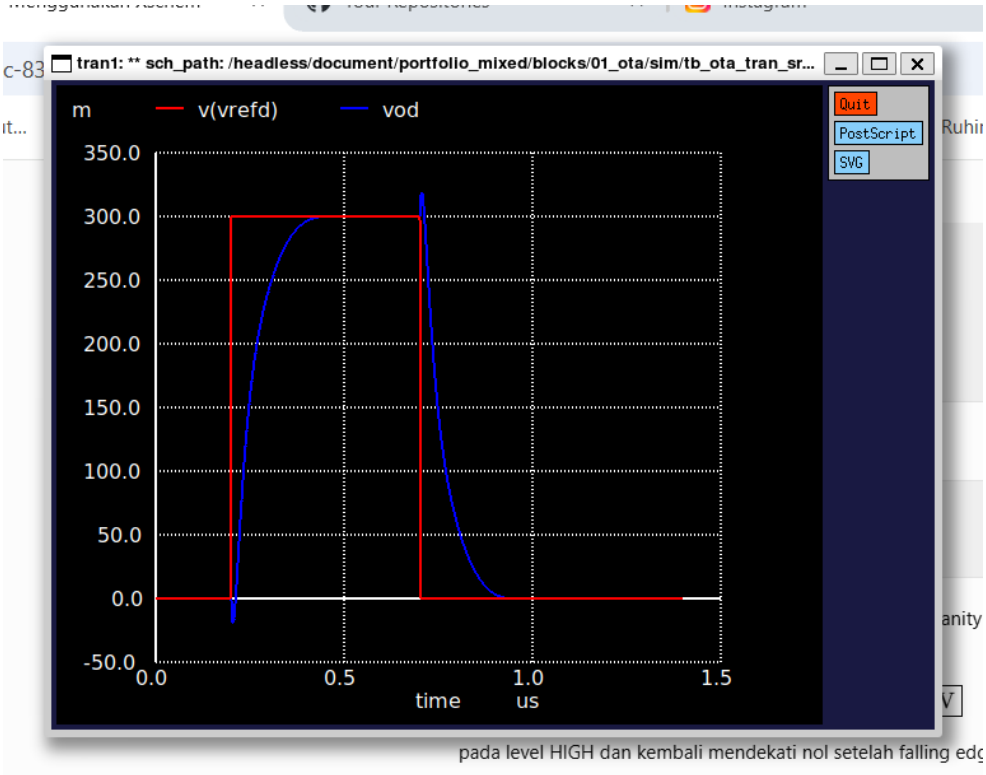


Figure 4: Large-signal differential response used for slew-rate characterization.

The measured peak differential slopes are

$$SR_+ \approx 5.059\,60\,\mathrm{V/\mu s}, \tag{7}$$

$$SR_- \approx 4.970\,22\,\mathrm{V/\mu s}. \tag{8}$$

The positive and negative values are closely matched. During the large-signal event, the output common mode ranged approximately from 0.861 V to 0.920 V, indicating a larger common-mode disturbance than in the small-signal settling test.

7 Quiescent Current and Power

At the nominal balanced input operating point, the main supply current is approximately

$$I_{DD} = 6.438\,659\,\mathrm{mA}. \tag{9}$$

The corresponding DC power is

$$P_{DC} = V_{DD} I_{DD} \approx 11.589\,59\,\mathrm{mW}. \tag{10}$$

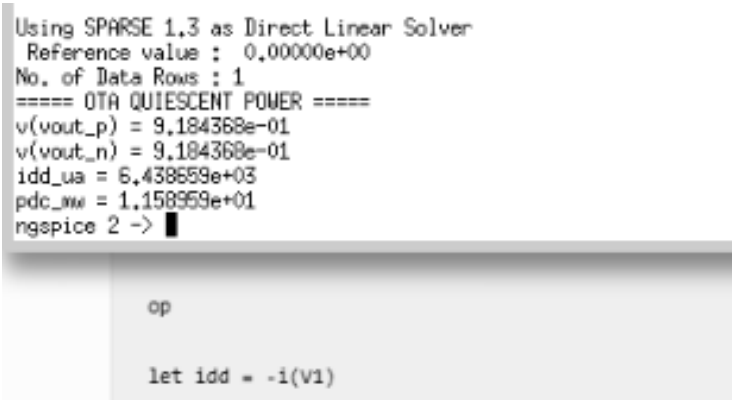


Figure 5: Nominal quiescent-current and DC-power measurement.

This power level is acceptable for a proof-of-concept analog portfolio block but is relatively high for a low-power converter target. The output-stage sizing and bias strategy should therefore be revisited if low-power operation becomes a primary specification.

8 Noise Characterization

Differential noise analysis was performed from 1 Hz to 25 kHz, matching the nominal baseband implied by 4 MHz sampling and OSR 80. A 1-V AC differential test source was used so ngspice could calculate both output noise and input-referred noise.

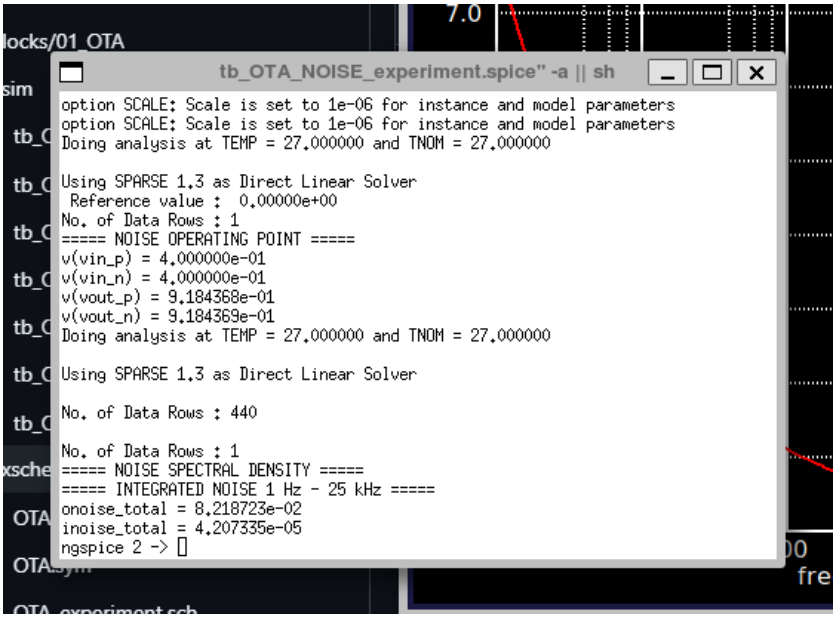


Figure 6: Noise operating point and integrated-noise result.

The integrated results are

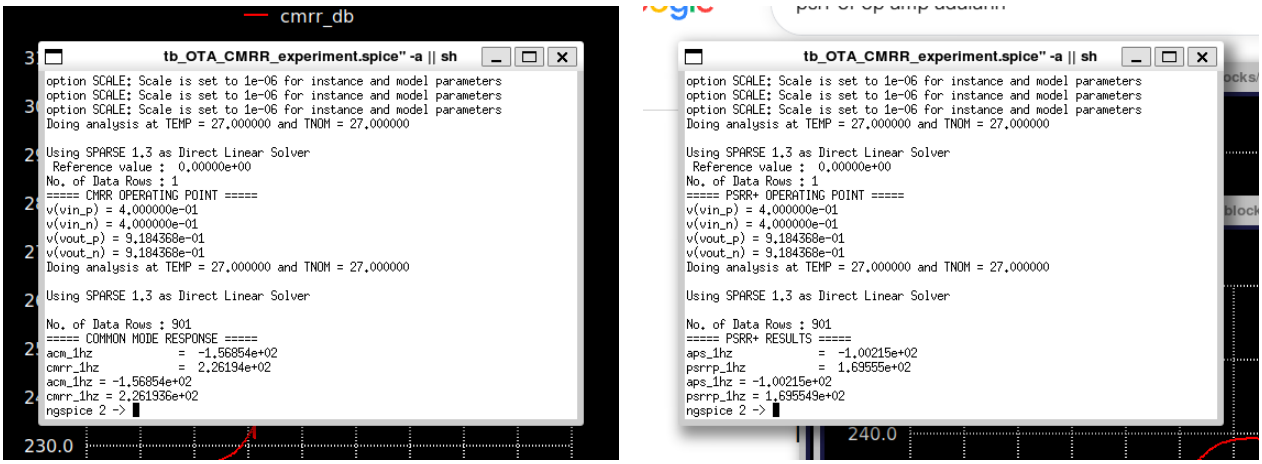
$$v_{n,out,rms} \approx 82.187\,23\,\mathrm{mV_{rms}}, \tag{11}$$

$$v_{n,in,rms} \approx 42.073\,35\,\mu\mathrm{V_{rms}}. \tag{12}$$

The input-referred value is the more useful block-level metric. A final converter-level noise budget should compare this figure with the switched-capacitor kT/C contribution, quantization noise, signal full scale, and required dynamic range.

9 CMRR and PSRR

Common-mode rejection was evaluated by applying identical AC excitation to both inputs and observing the differential output. At 1 Hz, the simulated common-mode gain is approximately −156.854 dB. Relative to the nominal differential gain, this produces a calculated nominal CMRR of approximately 226.194 dB.



(a) Nominal CMRR result. (b) Nominal PSRR+ result.

Figure 7: Rejection measurements at the nominal schematic level.

For positive-supply rejection, the supply-to-differential-output coupling at 1 Hz is approximately −100.215 dB, corresponding to a nominal calculated PSRR+ of approximately 169.555 dB. For PSRR-, ideal schematic symmetry drove the differential coupling below the explicit numerical floor used to avoid log(0); therefore no finite physical PSRR- value is reported.

The CMRR and PSRR values above are **nominal schematic symmetry checks**, not silicon predictions. Device mismatch, bias coupling, interconnect parasitics, asymmetrical loading, and layout gradients will reduce the achievable rejection. Monte Carlo and post-layout extraction are required before these metrics can be considered implementation-level results.

10 PVT Stress Characterization

10.1 Method

PVT testing deliberately retained the fixed nominal bias $V_{bias} = 0.507\,621$ V so that sensitivity of the complete OTA bias point could be observed directly. Seven stress points were run: nominal TT, two supply extremes, two temperature extremes, and SS/FF process corners at nominal supply and temperature.

A corner was marked **FAIL** if the DC output common mode collapsed near a rail or if the amplifier no longer exhibited meaningful positive low-frequency gain. In failed operating points, numerically reported UGF or phase values are not interpreted as valid amplifier metrics.

Table 4: PVT stress results with fixed nominal bias.

Process	V_{DD} (V)	Temp. (°C)	V_{OCM} (V)	A_0 (dB)	I_{DD}	Power	Status / observation
TT	1.80	27	0.9184	69.34	6.439 mA	11.590 mW	**PASS**
TT	1.62	27	1.6129	-57.23	16.7 nA	27.1 nW	**FAIL**; upper-rail collapse
TT	1.98	27	0.1306	-103.04	14.44 mA	28.58 mW	**FAIL**; lower-rail collapse
TT	1.80	-40	1.7978	-61.26	31.3 nA	56.3 nW	**FAIL**; upper-rail collapse
TT	1.80	125	0.1450	-82.86	10.02 mA	18.04 mW	**FAIL**; lower-rail collapse
SS	1.80	27	0.1010	-90.19	9.513 mA	17.12 mW	**FAIL**; lower-rail collapse
FF	1.80	27	1.7892	-4.04	0.810 μA	1.459 μW	**FAIL**; upper-rail collapse



(a) TT, -40°C (b) TT, 125°C (c) TT, 1.62 V

(d) TT, 1.98 V (e) SS, 1.8 V (f) FF, 1.8 V

Figure 8: Representative failed fixed-bias PVT operating points.

10.2 Bias-sweep diagnostic

A separate temperature-bias sweep was used as a diagnostic exercise. At TT / 1.8 V, a DC sweep found approximately 0.470 423 V at -40°C and 0.591 558 V at 125°C for a sweep crossing near the nominal output common-mode level. However, the cold-point crossing did not reproduce as a robust standalone AC operating point: a standalone solve returned a different equilibrium. This path dependence indicates that simply replacing the fixed voltage with two manually tuned values is not sufficient evidence of PVT robustness.

The correct design implication is therefore not “retune V_{bias} at every corner”, but rather to implement a physically meaningful bias/current-reference network with controlled PVT behavior and then repeat the full characterization.

11 Revision 2 Addendum: Compensation Sensitivity and Integration Notes

This addendum records two checks made after the block was frozen. It changes no nominal metric of the preceding sections. Status of all results below: measured, not final (OTA alone, ideal common-mode control, 20 pF per side, TT / 1.8 V / 27 °C).

11.1 Compensation capacitance: MF=80 versus MF=100

The AC testbench of Section 4 was reused unchanged. The only difference between the two runs is a copy of the OTA schematic in which `MF` of both compensation capacitors was changed from 80 to 100. The `MF=80` run reproduces the Block-1 values (gain, UGF and phase margin) to the printed digits.

Table 5: Effect of the compensation capacitance (all other conditions identical).

Metric	MF=80 (211.6 fF)	MF=100 (264.5 fF)	Change
DC gain A_0	69.34 dB	69.34 dB	0.00 dB
UGF	3.1531 MHz	2.7111 MHz	−14.0%
Phase at UGF	−118.63°	−113.39°	
Phase margin	61.37°	66.61°	+5.24°
Gain at 100 kHz	30.03 dB	28.65 dB	
Gain at 1 MHz	10.02 dB	8.65 dB	

A $1/C_c$ scaling of the UGF would predict a −20% change for this 25% increase of capacitance; the measured change is −14%, which corresponds to an empirical exponent of about −0.68 between these two points. The UGF is therefore not set by C_c alone.

Two further observations from the same harness: (i) raising V_{bz} from 1.5 V to 1.8 V gave results identical to the printed precision, including the DC operating point; the reason was not investigated. (ii) Setting $V_{bias} = 0.58$ V drove both outputs to about 1.79 V (gain −38.5 dB, no unity-gain crossing). In this harness the control voltage V_{cmfb1} is a fixed source, so the output common mode is not regulated; that run is a limitation of the harness and is not a characterization of the OTA at that bias.

11.2 Relation to the OTA used in the modulator netlists

The transistor-level modulator netlists (subcircuits `S1_OTA_CMFB` and `S2_OTA_CMFB`, equivalent to the repository schematic `OTA_CMFB`) use the same transistor naming as this block but *not* the sizing of `OTA_experiment`:

Table 6: Sizing W/L in µm: characterized OTA versus modulator netlists.

Devices	`OTA_experiment` (this report)	Modulator netlists
M1, M2 (PMOS)	120 / 1.2	240 / 0.6
M3, M4 (NMOS)	8 / 2	8 / 2
M5, M6 (NMOS)	20 / 2	40 / 2
M7, M8 (PMOS)	30 / 2	60 / 2
M9, M10 (PMOS)	300 / 1	60 / 1
M11, M12 (NMOS)	200 / 1	80 / 1
M13 (PMOS)	60 / 1.2	480 / 1.2
V_{bz} series branches (NMOS 5/5)	removed	present, in series with C_c
Compensation capacitors	`MF=80` (211.6 fF)	`MF=100` (264.5 fF)

The repository schematic `OTA_CMFB` carries `MF=1000`; the generated modulator netlists use `MF=100`. The metrics of this report therefore describe `OTA_experiment` only. They have not been measured for the OTA instantiated in the modulator netlists, and the closed-loop result obtained with those netlists (Block 5 report) is not evidence for the metrics reported here. AC characterization of the modulator OTA with its switched-capacitor common-mode feedback closed, at its loop bias, remains an open item.

12 Final Performance Summary

Table 7: Final nominal Block-1 OTA metrics.

Metric	Result	Interpretation
DC gain	69.34 dB	Valid nominal
UGF	3.15312 MHz	Valid nominal
Phase margin	61.3704°	Stable nominal loop
Rise time	56.1867 ns	50-mV closed-loop step
Settling, 1%	186.859 ns	50-mV closed-loop step
Settling, 0.1%	294.795 ns	50-mV closed-loop step
Overshoot	3.57043%	Small-signal step
SR_+ / SR_-	5.0596 / 4.9702 V/μs	300-mV step
Quiescent current	6.438659 mA	TT / 1.8 V / 27°C
DC power	11.58959 mW	TT / 1.8 V / 27°C
Input-referred noise	42.07335 μV_{rms}	1 Hz–25 kHz
Output noise	82.18723 mV_{rms}	1 Hz–25 kHz
CMRR @ 1 Hz	226.194 dB	Nominal, symmetry-limited
PSRR+ @ 1 Hz	169.555 dB	Nominal, symmetry-sensitive
PSRR- @ 1 Hz	Numerical floor	No finite physical value claimed
PVT fixed-bias	Nominal point only passes	Bias redesign required

13 Design Assessment and Block Closure

The work in this block demonstrates an end-to-end analog design workflow rather than only a nominal AC plot. The OTA was debugged at operating-point level, compensation was iterated, closed-loop behavior was verified, large-signal response was measured, power and noise were quantified, rejection tests were performed, and the design was deliberately stressed outside the nominal point.

The final assessment is split into two statements:

1. **Nominal characterization complete.** At TT / 1.8 V / 27 °C, the selected direct-C OTA is a valid characterized block with measured gain, bandwidth, phase margin, settling, slew rate, power, noise, CMRR, and PSRR behavior.
2. **PVT qualification not achieved.** The fixed external bias voltage is not robust across the tested process, voltage, and temperature stress points. A bias/current-reference redesign is required before this OTA can be treated as tapeout-ready.

Accordingly, Block 1 can be frozen for the purpose of continuing system-level architecture and portfolio development, provided that the PVT limitation remains an explicit open implementation requirement. The next system block may use the nominal OTA model, while a future silicon-ready revision should incorporate PVT-aware biasing, Monte Carlo mismatch analysis, extracted parasitics, and layout-aware re-characterization.

A Reproducibility Notes

The companion package includes CSV files containing the final nominal metrics and the seven PVT stress points. The figures in this report are direct screenshots from the validated ngspice/Xschem runs. The characterization relied on separate testbenches for AC, closed-loop transient, slew rate, power, noise, CMRR, PSRR+, PSRR-, and PVT testing.

For future revision control, the recommended sign-off sequence is: regenerate the netlist, verify the intended MIM-capacitor multiplicity (done in Revision 2, Section 3.2), reconcile the OTA used in the modulator netlists with the characterized OTA (Section 11.2), rerun nominal characterization, rerun PVT with the redesigned bias network, then add mismatch Monte Carlo and post-layout extraction.